Modeling Stellar Spectra with The Cannon

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We implement and evaluate The Cannon (Ness et al. 2015), a data-driven spectral model, using APOGEE DR17 H-band spectra of red giant stars. The model is a 2nd-order polynomial in five stellar labels ($T_{\rm eff}$, $\log g$, [Fe/H], [Mg/Fe], [Si/Fe]) with 21 coefficients and one intrinsic scatter term per pixel, totaling 188,650 free parameters across 8575 pixels. Cross-validation on a held-out 50% sample recovers ASPCAP labels with scatter comparable to the pipeline uncertainties. MCMC fitting of a mystery spectrum yields precise posterior constraints on all five labels. We compare the Cannon to a neural network approach and discuss the tradeoffs between interpretability and flexibility.

1. Introduction

Stellar spectroscopy is the primary tool for measuring the physical properties of stars: effective temperature, surface gravity, and detailed chemical abundances. Traditional approaches rely on synthetic stellar atmosphere models to fit observed spectra, but these are computationally expensive and limited by the accuracy of atomic and molecular line lists.

The Cannon (**?**) offers a complementary, data-driven approach. Rather than computing spectra from first principles, it learns the mapping from stellar labels to spectra directly from a training set of well-characterized stars. The model is a per-pixel polynomial in the stellar labels, trained by maximizing the likelihood of the observed spectra. Once trained, it can rapidly predict spectra for new label combinations and, inversely, infer labels from observed spectra.

In this work, we apply The Cannon to APOGEE DR17 spectra of red giant stars in the H-band (15100–17000 Å). We train the model, validate it via cross-validation, fit a mystery spectrum using MCMC, and compare with a neural network approach.

2. Training Set

2.1. Data Acquisition

We query the APOGEE DR17 allStar catalog for four fields: M15 (metal-poor globular cluster, [Fe/H] ≈ -2.3), NGC 6791 (metal-rich open cluster, [Fe/H] $\approx +0.4$), K2_C4_168-21 (Kepler/K2 field), and 060+00 (bulge field). This provides a training set spanning a wide range of metallicities and evolutionary states. After quality cuts (all five ASPCAP labels finite and $\neq -9999$; SNR > 50; $\log g \leq 4$; $T_{\rm eff} \leq 5700$ K; [Fe/H] ≥ -1), the training set contains 1886 stars. Note that the [Fe/H] ≥ -1 cut removes most M15 members (whose cluster-mean metallicity is well below -1); only a handful of M15 stars with ASPCAP-reported [Fe/H] near -1 survive the cut.

Figures 1 and 2 show the distribution of stellar labels in the training set. The field-colored corner plot confirms that M15 (metal-poor) and NGC 6791 (metal-rich) occupy distinct regions of label space, with tight intra-cluster label distributions demonstrating ASPCAP label consistency.

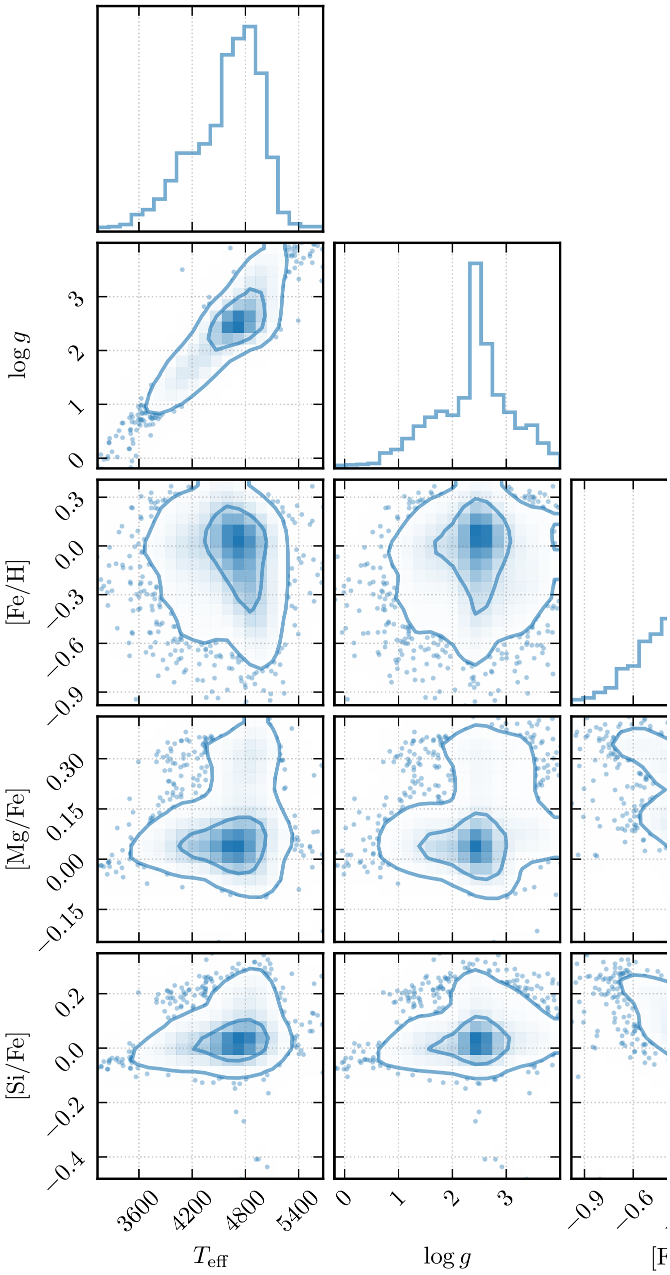

Fig. 1.— Corner plot of the five ASPCAP stellar labels in the training set, showing correlations between $T_{\rm eff}$, $\log g$, [Fe/H], [Mg/Fe], and [Si/Fe].

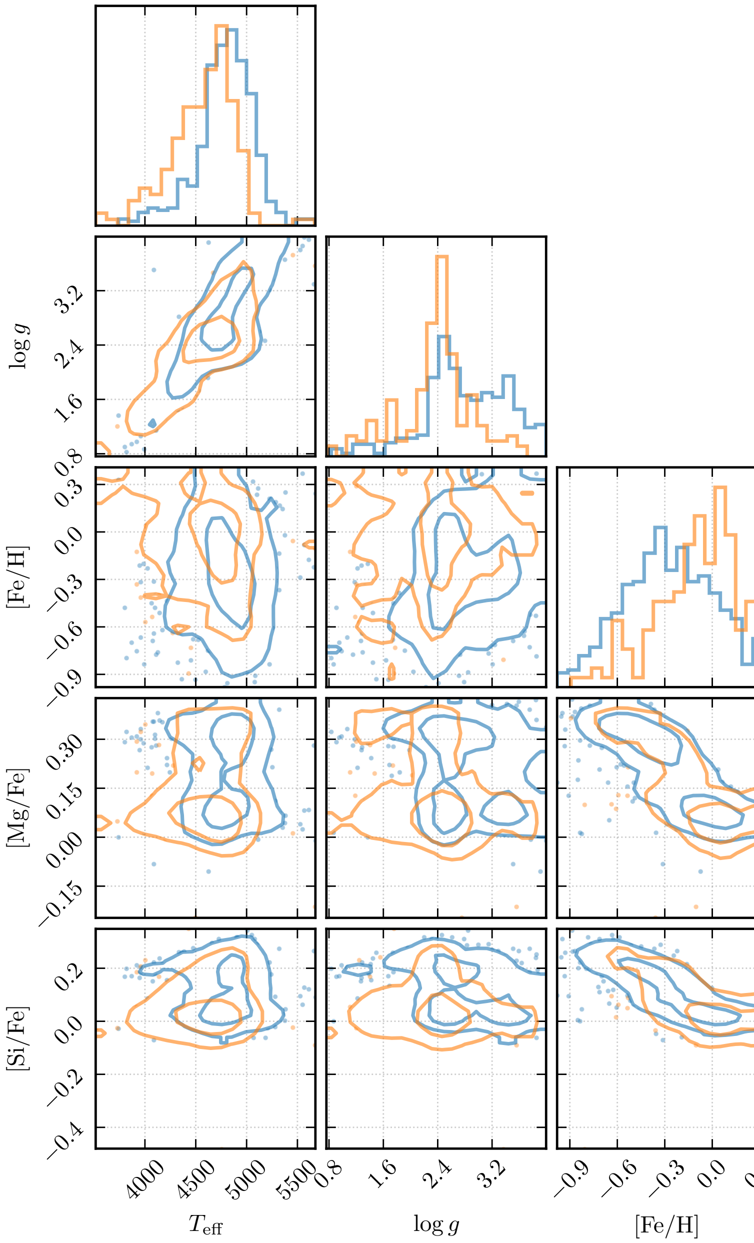

Fig. 2.— Corner plot colored by APOGEE field, showing the distinct label-space separation between M15 (metal-poor, α-enhanced) and NGC 6791 (metal-rich, solar α).

2.2. Surface Gravity

Surface gravity is $g = GM/R^2$. For a solar-mass star at different evolutionary stages: main sequence ($R \approx 1\,R_\odot$, $\log g \approx 4.44$), tip of the RGB ($R \approx 100\,R_\odot$, $\log g \approx 0.44$), and red clump ($R \approx 15\,R_\odot$, $\log g \approx 2.08$). The $\log g \leq 4$ cut selects giants.

2.3. Spectra and Normalization

APOGEE spectra span 15100–17000 Å across three detector chips (blue, green, red) with 8575 pixels on a log-linear wavelength grid. Raw spectra are in units of $10^{-17}\,\mathrm{erg\,s^{-1}\,cm^{-2}\,\AA^{-1}}$, i.e. spectral flux density per unit wavelength. The coadded spectra have been Doppler-shifted to the barycentric rest frame — the frame of the Solar System barycenter — to remove the Earth's orbital and rotational velocity components, which differ for each visit due to the changing line-of-sight velocity projection throughout the year. Figure 3 shows a representative raw spectrum.

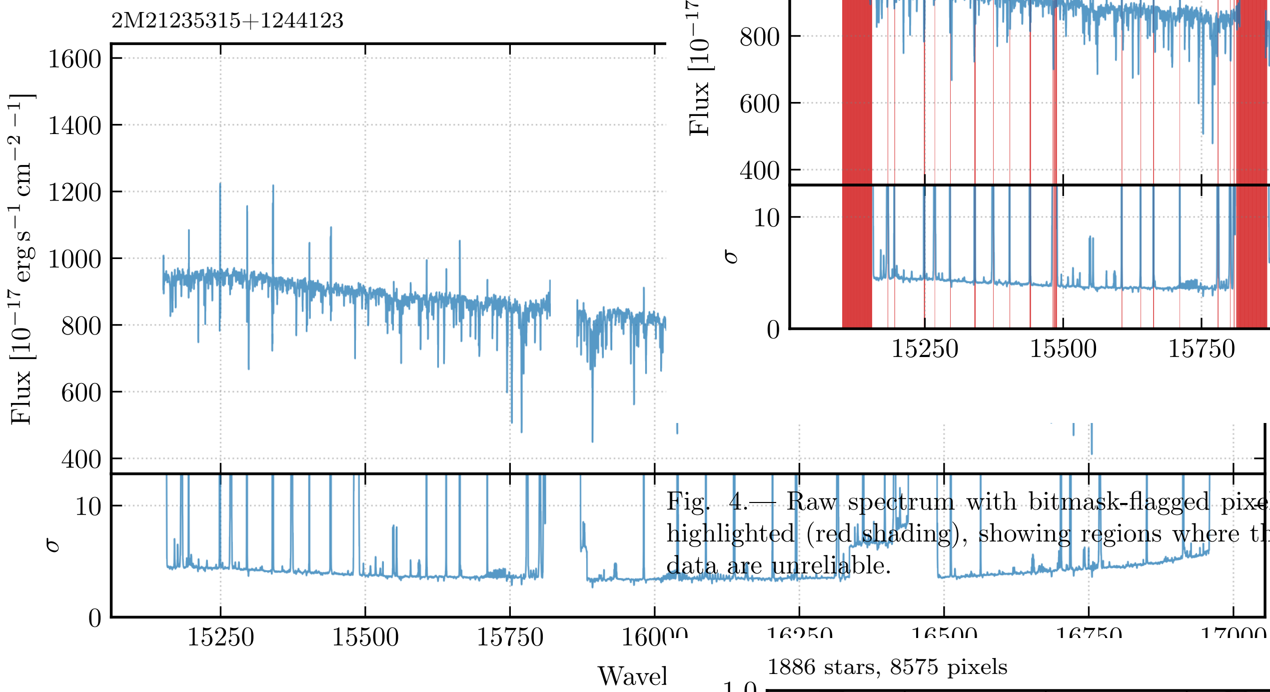

Fig. 3.— Raw APOGEE spectrum before normalization, showing the broadband SED shape and absorption features.

Pixels flagged by the APOGEE bitmask (bits 0–7 for detector issues and bit 12 for unvisited pixels) have their errors set to a large sentinel value (10^6), giving them negligible weight ($\propto 1/\sigma^2$) in subsequent fits. Negative-flux pixels (unflagged cosmic rays or artifacts) are set to NaN. Figure 4 illustrates the affected pixels for an example star, and Figure 5 shows the per-pixel flagging frequency across the training set. The dominant bad-pixel sources are the inter-chip gaps (∼15130, 15840, 16460, 16975 Å), OH airglow (∼15852 Å), and telluric CO_2 absorption (∼16452–16468 Å).

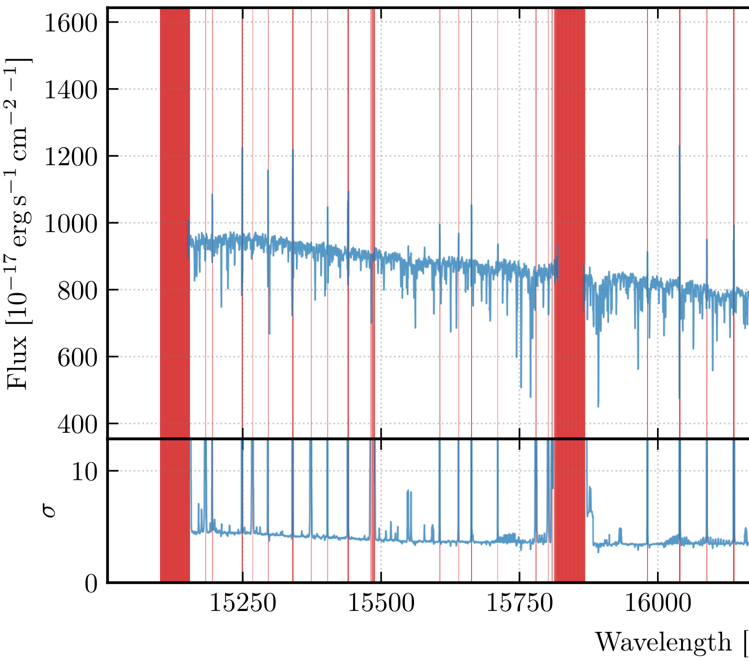

Fig. 4.— Raw spectrum with bitmask-flagged pixels highlighted (red shading), showing regions where the data are unreliable.

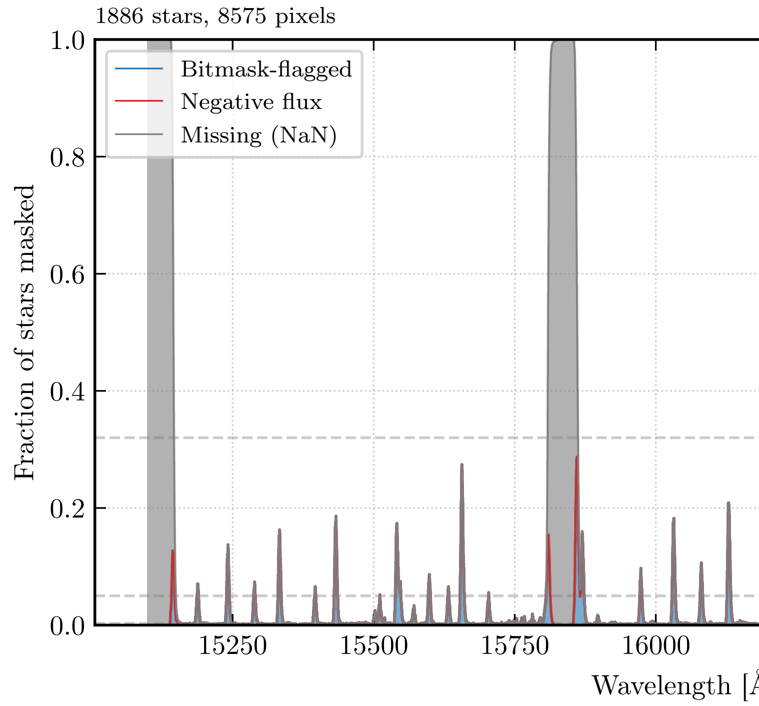

Fig. 5.— Per-pixel flagging frequency across the training set, stacked by cause (bitmask, negative flux, NaN). Horizontal guides mark the 68%, 95%, and 99.7% good-pixel thresholds.

We pseudo-continuum normalize each spectrum by fitting an independent degree-4 Chebyshev polynomial to 529 pre-identified continuum pixels per chip, then dividing. This is a *pseudo*-continuum because the "continuum" pixels are not truly line-free; they are simply wavelengths where the flux depends minimally on the stellar labels, as identified by Ness et al. (2015). True continuum normalization would require knowing the intrinsic stellar SED, which depends on the very labels we aim to measure. The pseudo-continuum approach removes the broadband shape while preserving the relative absorption-line depths that encode label information. Figure 6 shows this process.

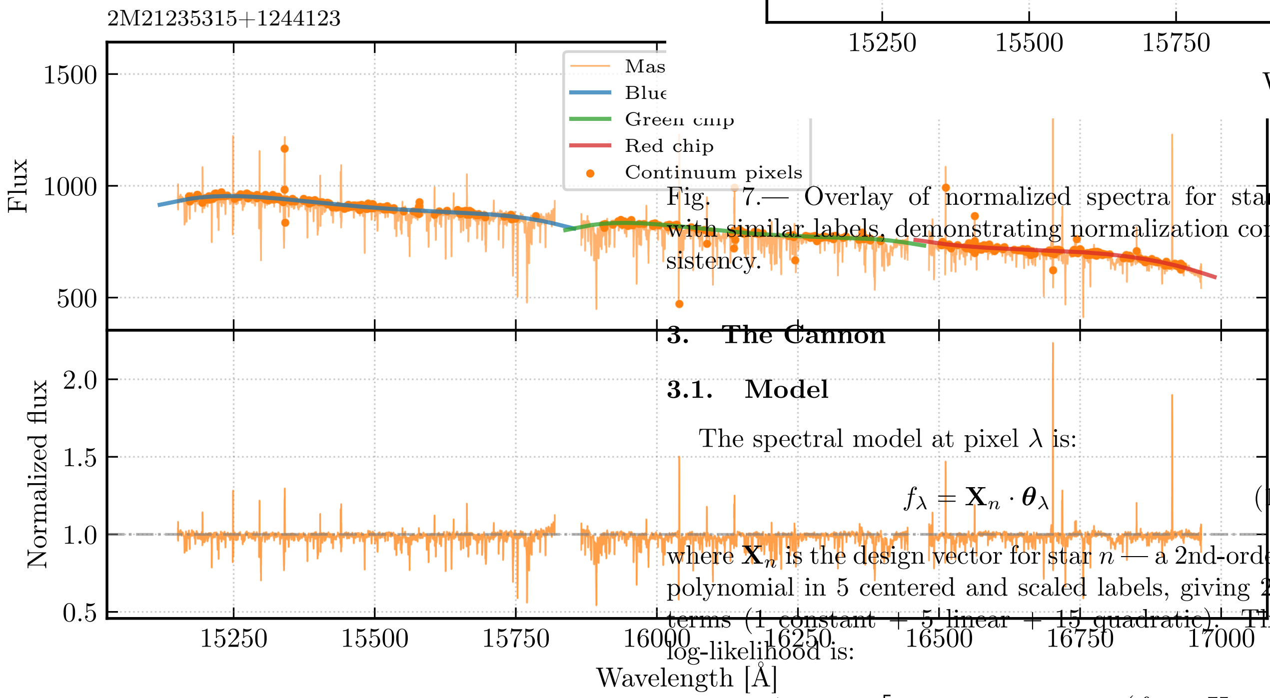

Fig. 6.— Normalization diagnostic for a single star: raw flux with per-chip Chebyshev continuum fit overlaid (top) and normalized spectrum (bottom). Continuum pixels used for fitting are marked.

Stars with similar labels produce nearly identical normalized spectra, confirming the normalization is robust (Figure 7).

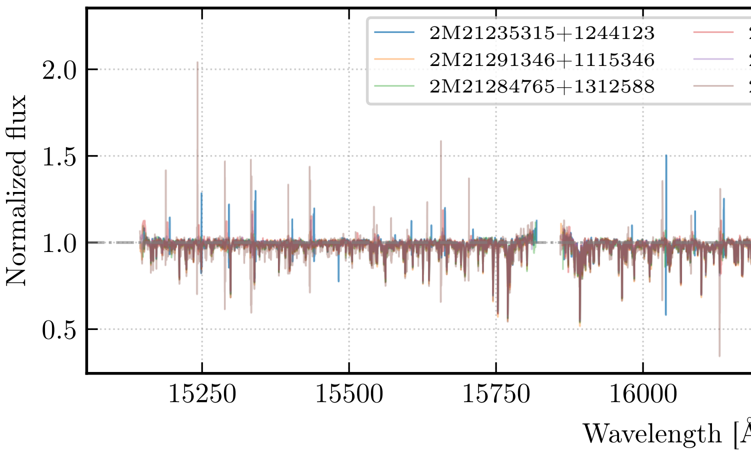

Fig. 7.— Overlay of normalized spectra for stars with similar labels, demonstrating normalization consistency.

3. The Cannon

3.1. Model

The spectral model at pixel λ is:

$$f_\lambda = \mathbf{X}_n \cdot \boldsymbol{\theta}_\lambda \tag{1}$$

where $\mathbf{X}_n$ is the design vector for star n — a 2nd-order polynomial in 5 centered and scaled labels, giving 21 terms (1 constant + 5 linear + 15 quadratic). The log-likelihood is:

$$\ln \mathcal{L} = -\frac{1}{2} \sum_n \sum_\lambda \left[\ln\left(2\pi(\sigma_{n\lambda}^2 + s_\lambda^2)\right) + \frac{(f_{n\lambda} - \mathbf{X}_n \cdot \boldsymbol{\theta}_\lambda)^2}{\sigma_{n\lambda}^2 + s_\lambda^2} \right] \tag{2}$$

where s_λ^2 is the per-pixel intrinsic scatter. We optimize via the profile likelihood: for fixed s_λ^2, $\boldsymbol{\theta}_\lambda$ is a weighted least-squares solution; we optimize the scalar $\log s_\lambda^2$ via bounded 1D minimization.

The total number of free parameters is $(21 + 1) \times 8575 = 188{,}650$.

3.2. Training Validation

Figure 8 compares the Cannon prediction against the observed spectrum for a training star in the 16000–16100 Å window.

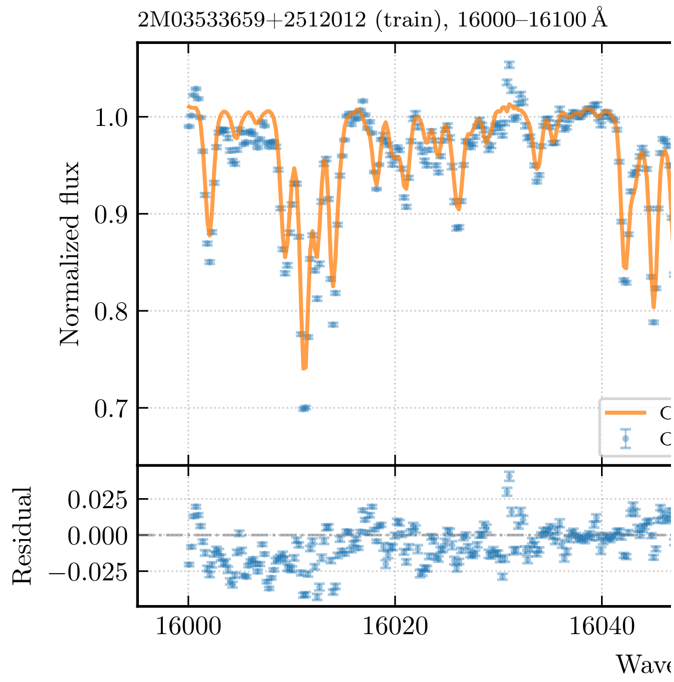

Fig. 8.— Observed vs. Cannon-predicted spectrum for a training star in the 16000–16100 Å window, with residuals.

3.3. Gradient Spectra

The gradient spectrum $\partial f_\lambda / \partial \ell_i$ reveals which spectral features are most sensitive to each label. Figure 9 shows the five gradient spectra with known Mg I and Si I lines marked. The $T_{\rm eff}$ gradient is large and broadband — essentially every pixel contributes information about temperature because the Boltzmann/Saha ionization balance affects all absorption features. This is why $T_{\rm eff}$ is the best-constrained label. In contrast, the abundance gradients ([Fe/H], [Mg/Fe], [Si/Fe]) are highly localized at specific metal-line wavelengths: the [Mg/Fe] gradient peaks sharply at the known Mg I triplet near 15750 Å, and the [Si/Fe] gradient peaks at Si I lines near 15893, 16064, and 16099 Å. The $\log g$ gradient is intermediate — concentrated at pressure-sensitive features (Brackett hydrogen line wings, strong neutral metal cores) but weaker than the $T_{\rm eff}$ gradient. This contrast between broadband temperature sensitivity and line-localized abundance sensitivity is what enables the Cannon to disentangle all five labels simultaneously.

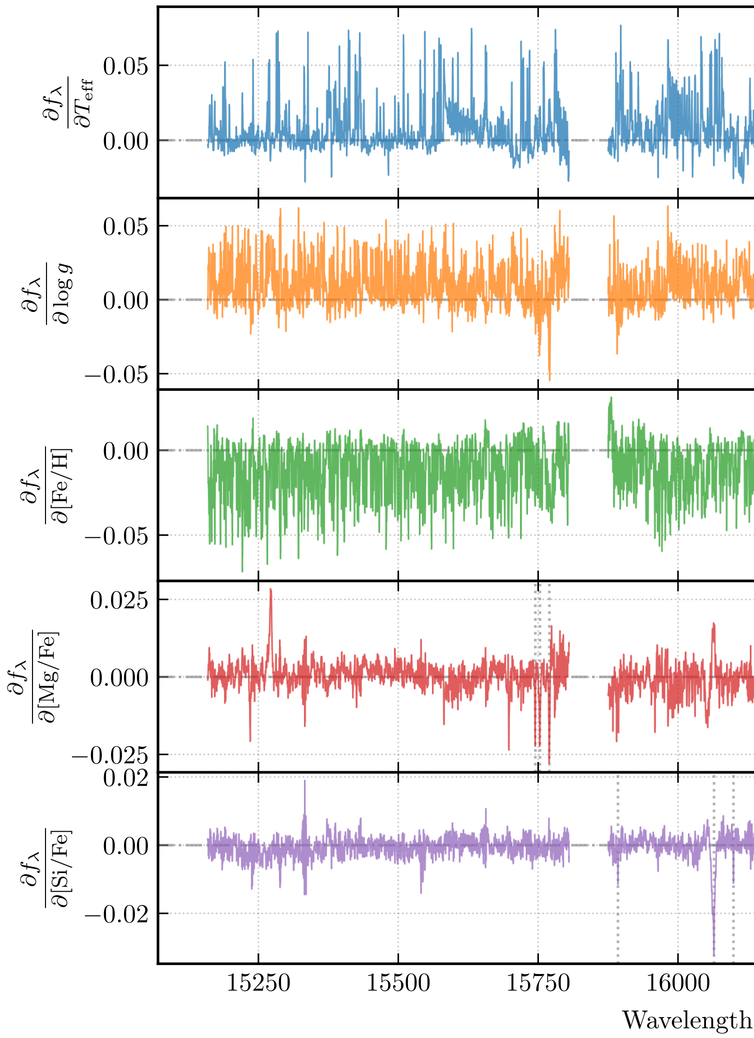

Fig. 9.— Gradient spectra for each of the five labels, evaluated at the training-set mean. Vertical lines mark known Mg I (on the [Mg/Fe] panel) and Si I lines (on the [Si/Fe] panel).

3.4. Intrinsic Scatter

Figure 10 shows the per-pixel intrinsic scatter s_λ. High-scatter regions fall into three categories: (1) *model inadequacy at strong absorption lines*, where the 2nd-order polynomial cannot capture the nonlinear flux response — this includes the Brackett hydrogen series (Br 12–Br 20, ∼15200–16400 Å), CO first-overtone bandheads (∼15582, 15780, 16398 Å), and saturated Fe I/Mg I/Si I lines; (2) *chip-edge artifacts* near ∼15800 and ∼16420 Å where marginal data quality at chip boundaries inflates scatter; and (3) *unflagged single-star outliers* producing isolated scatter spikes at individual wavelengths.

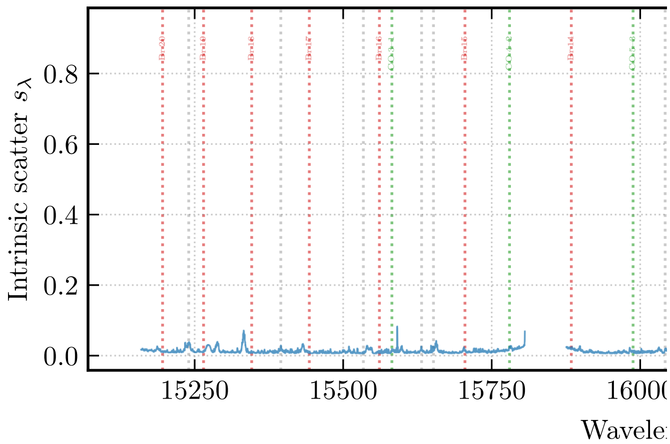

Fig. 10.— Per-pixel intrinsic scatter s_λ as a function of wavelength.

4. Cross-Validation

We split the training set 50/50 (seed = 40) and fit labels for the held-out CV stars using `scipy.optimize.lea` with analytical Jacobians. The analytical Jacobian $\partial r_\lambda/\partial \ell_j$ follows directly from the polynomial structure of the design vector, with chain-rule factors $1/\sigma_{\ell_j}$ from the label standardization. This allows convergence in 5–20 iterations rather than 50–200 with finite differences, and avoids the numerical noise that finite-difference Jacobians introduce when the step size is comparable to the label precision. Figure 11 shows the 1-to-1 recovery.

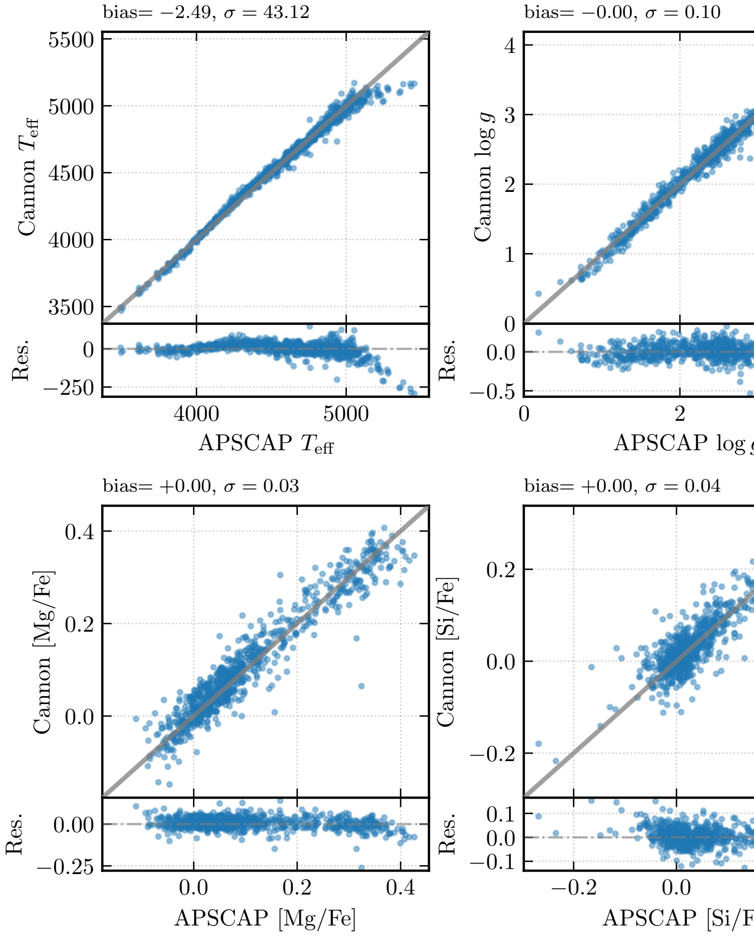

Fig. 11.— Cross-validation label recovery: fitted vs. true ASPCAP labels for each of the five labels, with residuals and bias/scatter annotations.

4.1. Outlier Investigation

Figure 12 shows observed vs. model spectra for the worst-fit CV stars. For each outlier, we check three potential failure modes: (1) whether the optimizer converged to a non-trivial solution (not stuck at the training-set mean), (2) whether the spectrum or continuum normalization has visible artifacts, and (3) whether the ASPCAP `starflag`/`aspcapflag` bitmasks indicate unreliable pipeline labels. In most cases the optimizer converges but the ASPCAP flags are non-zero, suggesting the reference labels themselves are unreliable rather than the Cannon fit being poor.

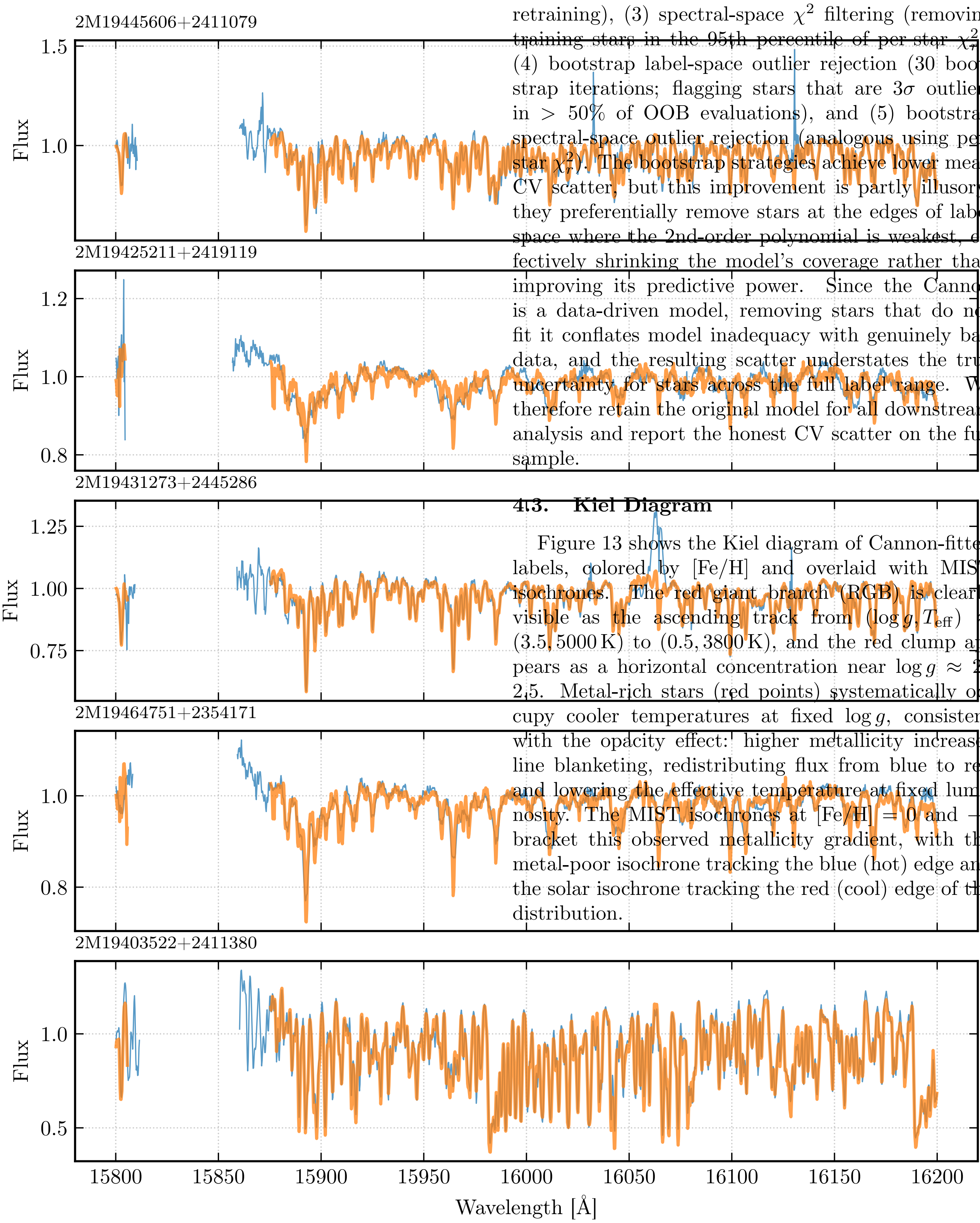

Fig. 12.— Observed and Cannon-predicted spectra for the worst-fit CV stars.

4.2. Model Selection

We explore five outlier-rejection strategies to understand the sensitivity of the CV scatter to training-set composition: (1) the original model with no changes, (2) label-space 3σ-clipping (removing CV stars whose residual exceeds 3σ in any label, then re-splitting and retraining), (3) spectral-space χ^2 filtering (removing training stars in the 95th percentile of per-star χ^2_r), (4) bootstrap label-space outlier rejection (30 bootstrap iterations; flagging stars that are 3σ outliers in $> 50\%$ of OOB evaluations), and (5) bootstrap spectral-space outlier rejection (analogous using per-star χ^2_r). The bootstrap strategies achieve lower mean CV scatter, but this improvement is partly illusory: they preferentially remove stars at the edges of label space where the 2nd-order polynomial is weakest, effectively shrinking the model's coverage rather than improving its predictive power. Since the Cannon is a data-driven model, removing stars that do not fit it conflates model inadequacy with genuinely bad data, and the resulting scatter understates the true uncertainty for stars across the full label range. We therefore retain the original model for all downstream analysis and report the honest CV scatter on the full sample.

4.3. Kiel Diagram

Figure 13 shows the Kiel diagram of Cannon-fitted labels, colored by [Fe/H] and overlaid with MIST isochrones. The red giant branch (RGB) is clearly visible as the ascending track from $(\log g, T_{\rm eff}) \approx (3.5, 5000\,{\rm K})$ to $(0.5, 3800\,{\rm K})$, and the red clump appears as a horizontal concentration near $\log g \approx 2$–2.5. Metal-rich stars (red points) systematically occupy cooler temperatures at fixed $\log g$, consistent with the opacity effect: higher metallicity increases line blanketing, redistributing flux from blue to red and lowering the effective temperature at fixed luminosity. The MIST isochrones at [Fe/H] $= 0$ and -1 bracket this observed metallicity gradient, with the metal-poor isochrone tracking the blue (hot) edge and the solar isochrone tracking the red (cool) edge of the distribution.

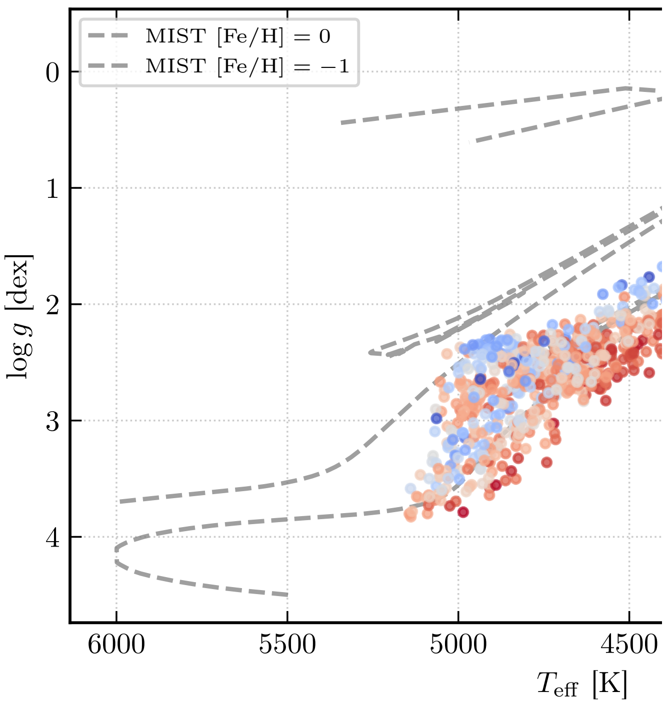

Fig. 13.— Kiel diagram ($\log g$ vs. $T_{\rm eff}$) of CV-set stars with Cannon-fitted labels, colored by [Fe/H]. MIST isochrones at [Fe/H] = 0 and −1 (6 Gyr) are overlaid.

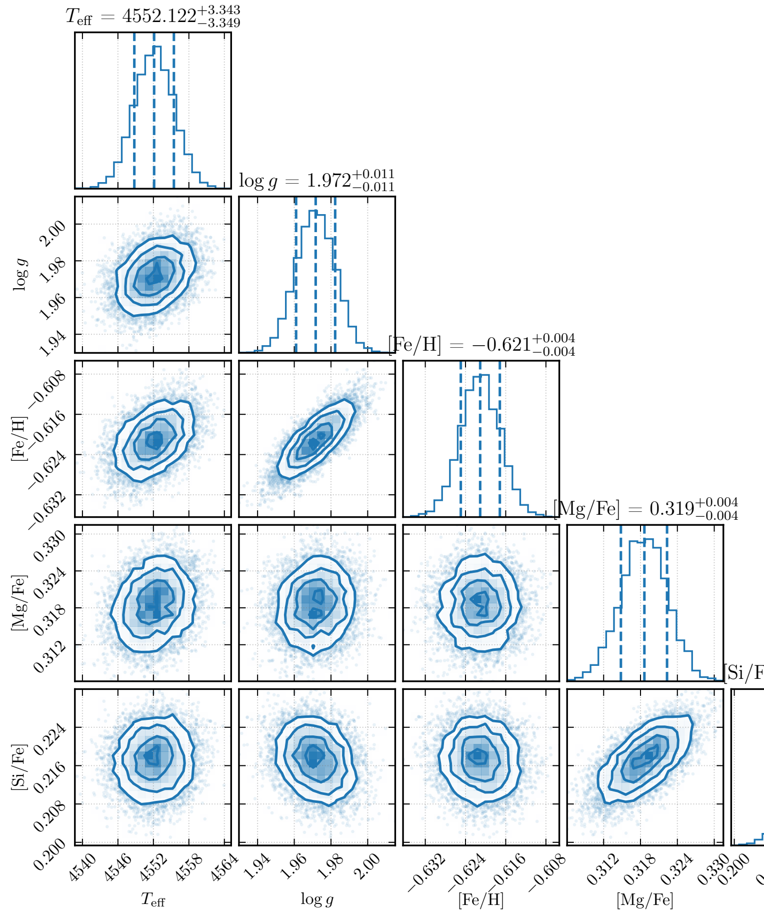

Fig. 14.— Corner plot of the MCMC posterior for the mystery spectrum, showing the joint and marginal distributions of the five stellar labels with 16th/50th/84th percentile quantiles.

5. MCMC Label Fitting

We fit labels for a mystery spectrum via MCMC (PyMC NUTS) using a `CannonLabelLikelihood` wrapper. The likelihood at each pixel is $f_\lambda^{\rm obs} \sim \mathcal{N}(f_\lambda^{\rm model}, \sigma_\lambda^2 + s_\lambda^2)$, where s_λ^2 is the Cannon's intrinsic scatter. Priors are weakly informative Normals centered on the training-set means with width 2σ. The posterior is sampled with 2000 draws after 1000 tuning steps (4 chains, seed 42). All chains converge: $\hat{R} < 1.01$ and effective sample size > 400 for every parameter. Figure 14 shows the posterior constraints.

Figure 15 compares the observed mystery spectrum against the Cannon prediction at the posterior median labels. The pull distribution (residual divided by $\sqrt{\sigma_\lambda^2 + s_\lambda^2}$) is centered near zero, confirming the fit is consistent with the noise model.

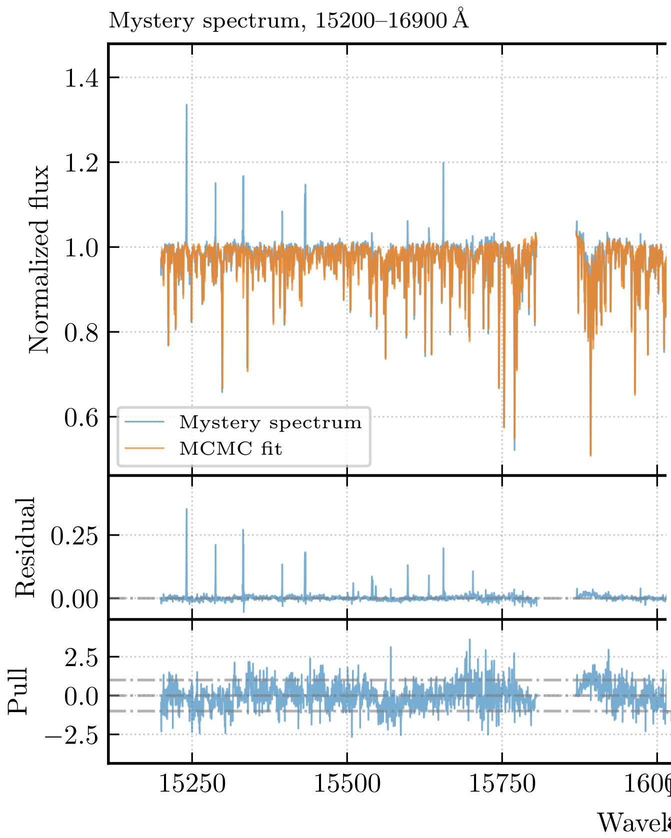

Fig. 15.— Mystery spectrum vs. MCMC-fitted Cannon prediction. Top: observed and model spectra. Middle: raw residuals. Bottom: pull (residual normalized by total uncertainty).

Formal MCMC uncertainties (e.g., ∼3 K in $T_{\rm eff}$) are an order of magnitude smaller than the CV scatter (∼56 K). This is expected: the MCMC likelihood treats pixels as independent Gaussians, neglecting (a) correlated continuum normalization residuals across neighboring pixels, (b) systematic line-blending effects where multiple labels influence the same spectral feature, and (c) model inadequacy that is only partially captured by the scalar s_λ^2 term. The CV scatter, which measures end-to-end recovery including all these systematics, provides a more realistic uncertainty estimate.

6. Spectral Synthesis

6.1. Metallicity Sequence

Figure 16 shows synthetic spectra at fixed $T_{\rm eff}$ = 4800 K, $\log g = 2.5$, varying [Fe/H] from −1 to +0.5.

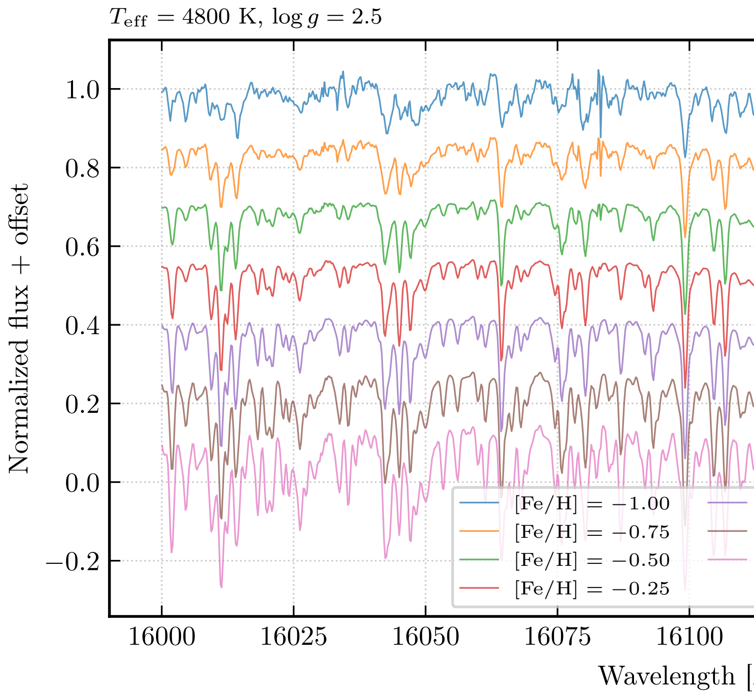

Fig. 16.— Synthetic Cannon spectra varying [Fe/H] at fixed $T_{\rm eff}$ and $\log g$, showing the effect of metallicity on absorption line strengths.

6.2. RGB Evolution

Figure 17 shows synthetic spectra along the RGB from a MIST isochrone ([Fe/H] = 0, $\log g$ from 3.5 to 0.5).

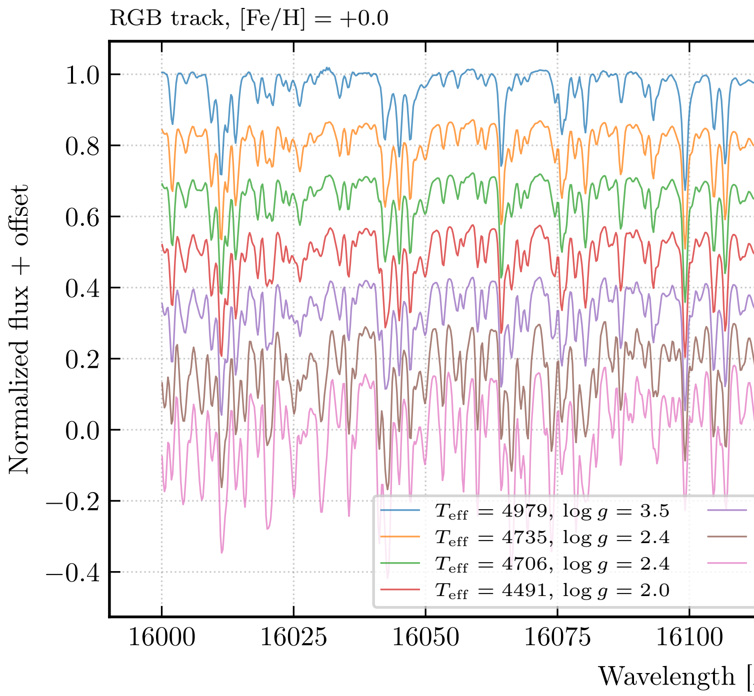

Fig. 17.— Synthetic Cannon spectra along the RGB evolutionary track at solar metallicity.

The metallicity sequence and RGB evolution produce distinguishable spectral changes: Mg/Si lines respond differently to abundance changes vs. temperature/gravity changes, allowing The Cannon to disentangle these effects.

6.3. Unresolved Binaries

An unresolved binary produces a composite spectrum that is the flux-weighted sum of two stellar spectra falling within the same fiber. When The Cannon fits a single-star model to this composite, the inferred labels are biased. Specifically: (a) $T_{\rm eff}$ is biased toward an intermediate value between the two components, with the bias magnitude depending on the flux ratio; (b) $\log g$ may be biased depending on the luminosity ratio, since the secondary's contribution dilutes the pressure-sensitive features; (c) abundances are biased toward zero because the secondary's continuum fills in the primary's absorption lines, reducing equivalent widths. The bias is largest for near-equal-mass binaries and diminishes for extreme mass ratios where the secondary contributes negligible flux. Possible mitigations include two-component spectral fitting (fitting two label vectors simultaneously), flagging candidates via radial-velocity scatter across visits, and spectral disentangling techniques (El-Badry et al. 2018).

7. Neural Network Comparison

We train a simple MLP ($8575 \to 512 \to 256 \to 5$) to predict labels from spectra. Figure 18 shows the training curves, and Figure 19 shows the CV label recovery.

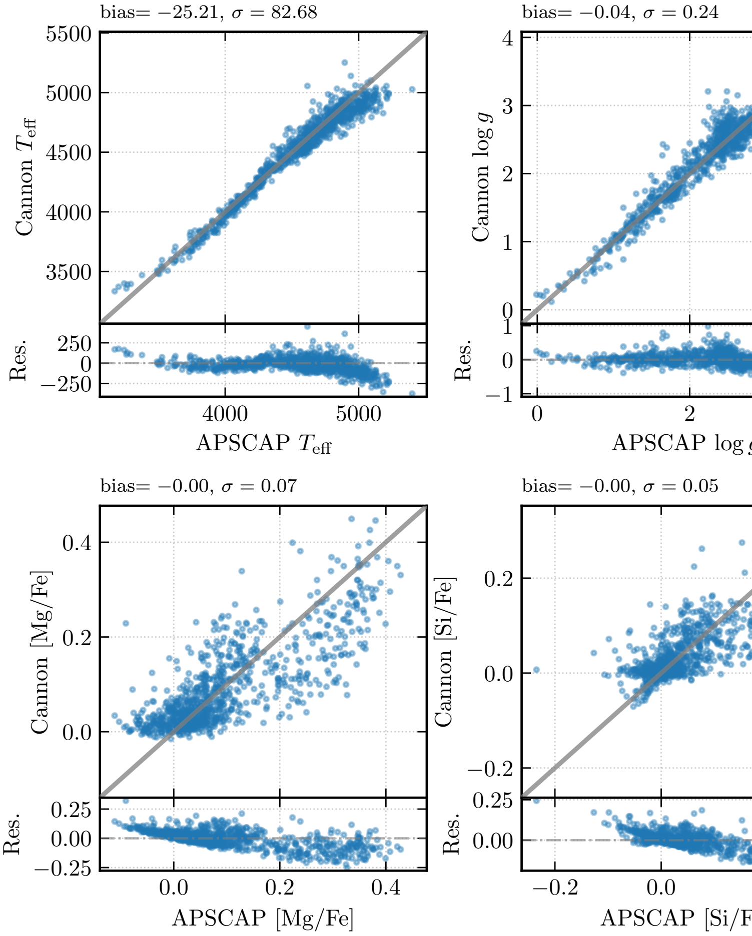

Fig. 19.— Neural network CV label recovery in the same format as Figure 11.

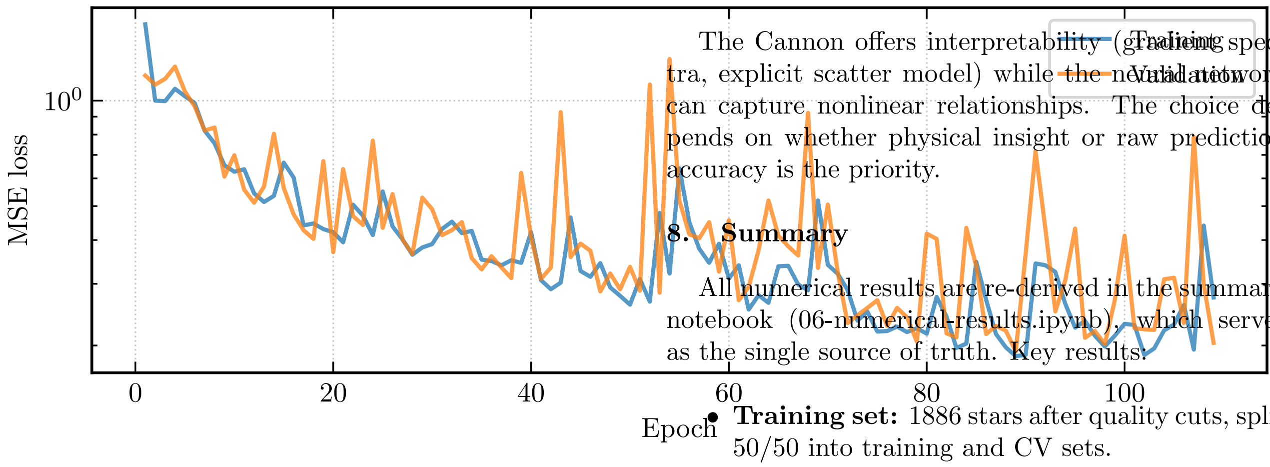

Fig. 18.— Training and validation MSE loss vs. epoch for the neural network.

The Cannon offers interpretability (gradient spectra, explicit scatter model) while the neural network can capture nonlinear relationships. The choice depends on whether physical insight or raw prediction accuracy is the priority.

8. Summary

All numerical results are re-derived in the summary notebook (06-numerical-results.ipynb), which serves as the single source of truth. Key results:

- **Training set:** 1886 stars after quality cuts, split 50/50 into training and CV sets.
- **Cannon model:** 21 coefficients + 1 scatter term per pixel × 8575 pixels = 188,650 free parameters. Training $\chi^2_r \approx 0.99$.
- **Cross-validation scatter:** $T_{\rm eff}$: ∼56 K; $\log g$: ∼0.14 dex; [Fe/H]: ∼0.04 dex; [Mg/Fe]: ∼0.05 dex; [Si/Fe]: ∼0.04 dex.
- **MCMC mystery star** (posterior medians): $T_{\rm eff} \approx 4552$ K, $\log g \approx 1.97$, [Fe/H] ≈ -0.62, [Mg/Fe] ≈ 0.32, [Si/Fe] ≈ 0.22.

- **Neural network:** MLP ($8575 \rightarrow 512 \rightarrow 256 \rightarrow 5$); CV scatter roughly 1.4–2.5$\times$ larger than The Cannon across all labels, as expected given the limited training set size relative to model capacity.